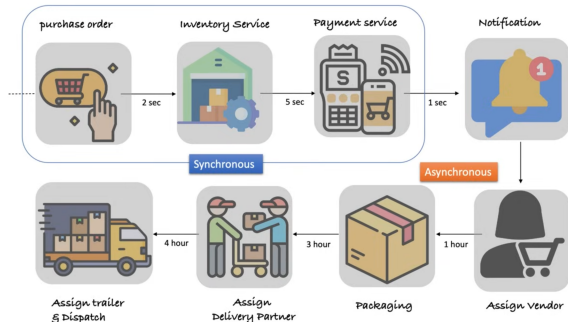


@Async – non block main Thread. Run task in separate threads. Till the payment I want sync flow and rest are all async flow.

<https://github.com/Java-Techie-jt/orderFulfillmentService-async>



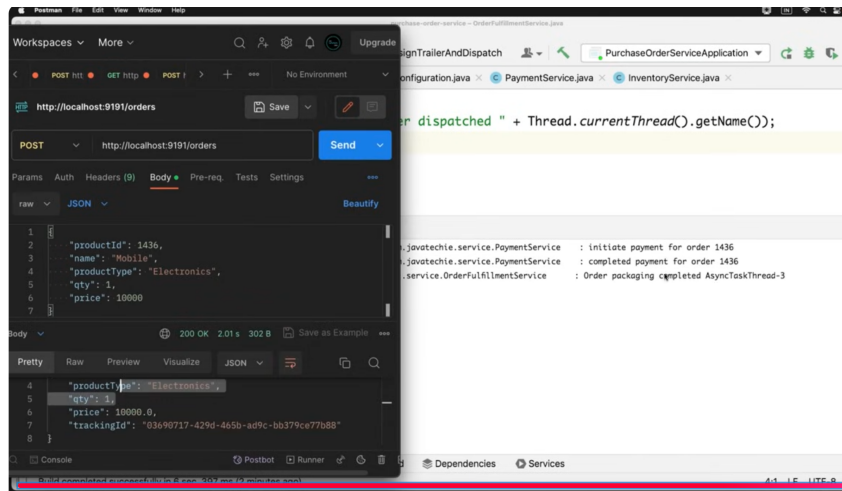
Bean config

```
8 import java.util.concurrent.Executor;
9
10 @Configuration
11 @EnableAsync
12 public class AsyncConfiguration {
13
14     @Bean("asyncTaskExecutor")
15     public Executor asyncTaskExecutor(){
16         ThreadPoolTaskExecutor taskExecutor=new ThreadPoolTaskExecutor();
17         taskExecutor.setCorePoolSize(4);
18         taskExecutor.setQueueCapacity(150);
19         taskExecutor.setMaxPoolSize(4);
20         taskExecutor.setThreadNamePrefix("AsyncTaskThread-");
21         taskExecutor.initialize();
22         return taskExecutor;
23     }
24 }
```

First one sync flow and rest are async flow

```
33 public Order processOrder(Order order) throws InterruptedException {
34     order.setTrackingId(UUID.randomUUID().toString());
35     if (inventoryService.checkProductAvailability(order.getProductId())) {
36         //handle exception here
37         paymentService.processPayment(order);
38     } else {
39         throw new RuntimeException("Technical issue please retry");
40     }
41     return order;
42 }
43
44 @Async("asyncTaskExecutor")
45 public void notifyUser(Order order) throws InterruptedException {
46     Thread.sleep(4000L);
47     log.info("Notified to the user " + Thread.currentThread().getName());
48 }
49
50 @Async("asyncTaskExecutor")
51 public void assignVendor(Order order) throws InterruptedException {
52     Thread.sleep(5000L);
53     log.info("Assign order to vendor " + Thread.currentThread().getName());
54 }
55
56 @Async("asyncTaskExecutor")
57 public void packaging(Order order) throws InterruptedException {
58     Thread.sleep(2000L);
59     log.info("Order packaging completed " + Thread.currentThread().getName());
60 }
61
62 @Async("asyncTaskExecutor")
63 public void assignDeliveryPartner(Order order) throws InterruptedException {
64     Thread.sleep(10000L);
65     log.info("Delivery partner assigned " + Thread.currentThread().getName());
66 }
67
68 @Async("asyncTaskExecutor")
69 public void assignTrailerAndDispatch(Order order) throws InterruptedException {
70     Thread.sleep(3000L);
71     log.info("Trailer assigned and Order dispatched " + Thread.currentThread().getName());
72 }
```

We got response for sync flow immediately. Rest are all running in separate thread without blocking main thread. That is why we got response in postman.



Even we got response in postman. Still background jobs are running. Notice logs in the above screen and below screen.

